

Notes: Khan and Thomas (Econometrica, 2008)
**Idiosyncratic Shocks and the Role of Nonconvexities in Plant and Aggregate
Investment Dynamics**

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1 Big Picture

One-sentence summary. This note summarizes Khan and Thomas (2008), which builds a quantitative general-equilibrium model of plant-level lumpy investment with persistent idiosyncratic productivity shocks and nonconvex adjustment costs, calibrates it to establishment-level investment-rate facts, and shows that nonconvexities can generate large aggregate nonlinearities in partial equilibrium but become quantitatively almost irrelevant once wages and interest rates adjust in general equilibrium.

Central question. Do plant-level nonconvex adjustment costs matter for aggregate investment dynamics once the model is disciplined by micro data and solved in general equilibrium?

Answer. The answer is subtle. Nonconvex adjustment costs matter a lot in partial equilibrium: they create skewness, fat tails, and strong extensive-margin responses in aggregate investment. But in general equilibrium, market-clearing price movements smooth the target-capital responses of plants, reduce the value of adjustment, and thereby remove most of the aggregate nonlinearity.

Core economic message. The paper does not deny that plant investment is lumpy. Instead, it argues that the macroeconomic consequences of lumpiness depend critically on equilibrium feedback. Micro frictions are not automatically macro frictions: equilibrium prices can absorb and neutralize the nonlinear plant-level responses.

2 Incremental Contribution of the Paper

1. **It reconciles micro realism with a general-equilibrium test.** Earlier general-equilibrium lumpy-investment models were criticized because they did not match the observed cross-sectional distribution of plant investment rates. This paper adds persistent plant-specific productivity heterogeneity and a band of low-level investment that avoids fixed costs, allowing the model to match key establishment-level investment facts.
2. **It separates partial-equilibrium from general-equilibrium implications.** The paper shows that the same micro model can produce strong aggregate nonlinearities under fixed factor prices but almost none under market-clearing wages and interest rates.
3. **It changes the interpretation of nonconvex costs.** In a model with large persistent idiosyncratic shocks, nonconvex costs do not mainly create investment spikes; instead, they eliminate some spikes that would otherwise arise from plant-specific productivity movements.
4. **It warns against estimating adjustment costs in partial equilibrium.** Because general-equilibrium price movements sharply dampen plant responses to aggregate shocks, panel-based estimates that ignore equilibrium feedback may overstate fixed adjustment costs.
5. **It provides a bridge between plant-level investment evidence and macro business-cycle modeling.** The paper directly addresses the claim that lumpy plant investment is a missing essential feature of standard aggregate models.

3 Economic Environment

3.1 Plants and production

There is a continuum of establishments. Each plant produces output using predetermined capital k and labor n :

$$y = z\varepsilon F(k, n),$$

where z is aggregate productivity and ε is persistent plant-specific productivity. The plant-level state is therefore not only capital but also idiosyncratic productivity.

Why this matters. If capital is the only source of plant heterogeneity, the model cannot match the observed distribution of plant investment rates. Persistent productivity heterogeneity generates a wider and more realistic distribution of desired capital stocks.

3.2 Capital adjustment and nonconvex costs

Capital evolves as

$$\gamma k' = (1 - \delta)k + i,$$

where γ is the growth factor and δ is depreciation. A plant can adjust freely within a small investment band, but to undertake a large adjustment it must pay a fixed labor cost. Specifically:

- small investment rates in $[a, b]$ can be undertaken without paying the fixed cost;
- larger positive or negative adjustments require paying a fixed cost ξ ;
- ξ is drawn from a distribution each period, so plants with the same (ε, k) may differ in whether adjustment is worthwhile.

This creates a generalized two-sided (S, s) investment rule. Plants compare the value of moving to their target capital with the value of remaining inside the constrained adjustment band.

3.3 Households and equilibrium prices

Households choose consumption, labor, and asset holdings. Equilibrium requires goods-market and labor-market clearing. As a result, the wage and discount factor faced by plants are not fixed; they move with aggregate states. These price movements are the key mechanism behind the paper's macro result.

4 Decision Rules and Economic Intuition

The plant has two possible capital choices:

- an **unconstrained target capital** $k^*(\varepsilon, z, \mu)$, chosen after paying the fixed cost;
- a **constrained capital choice** $k^c(\varepsilon, k, z, \mu)$, chosen within the no-cost adjustment band.

A plant pays the fixed cost if the gain from reaching the target capital is large enough. This creates a threshold rule:

$$\xi \leq \xi^T(\varepsilon, k; z, \mu).$$

Therefore, for each plant type (ε, k) , the fraction of adjusting plants is $G(\xi^T(\varepsilon, k; z, \mu))$.

Two margins matter.

1. **Intensive margin:** how far adjusting plants move their capital toward the target.
2. **Extensive margin:** how many plants choose to pay the fixed cost and adjust.

The lumpy-investment literature emphasizes that aggregate shocks can synchronize plant adjustment through the extensive margin. Khan and Thomas show that general-equilibrium price movements sharply dampen this synchronization.

5 Calibration and Experiments

The model is calibrated at an annual frequency. Common parameters are chosen to match standard postwar U.S. aggregate moments, while idiosyncratic productivity and adjustment-cost parameters are disciplined by the establishment-level investment-rate distribution.

The paper compares several economies:

- **Frictionless model:** no fixed adjustment costs; plants adjust every period.
- **Lumpy investment model:** fixed adjustment costs and constrained low-level investment band.
- **Common-productivity lumpy model:** removes plant-specific productivity heterogeneity to isolate the role of common shocks and capital nonconvexities.
- **Partial-equilibrium versions:** real wages and interest rates fixed at steady-state values.
- **General-equilibrium versions:** wages and discount factors adjust to clear markets.

6 Main Results

6.1 Result 1: The calibrated model matches plant-level investment facts

The model reproduces important features of the establishment investment-rate distribution, especially once it includes both persistent plant-specific productivity and low-level investment exempt from adjustment costs. These features allow the model to match inaction, small investment, positive spikes, negative investment, and asymmetry between positive and negative spikes.

6.2 Result 2: Nonconvexities create strong aggregate nonlinearities in partial equilibrium

When factor prices are fixed, a positive aggregate productivity shock raises target capital substantially. Many plants cross their adjustment thresholds at the same time, generating a large extensive-margin response. This produces skewness and fat tails in aggregate investment rates.

6.3 Result 3: General equilibrium nearly eliminates these nonlinearities

In general equilibrium, households smooth consumption. The wage and intertemporal price movements that support market clearing restrain plants' desired target-capital changes. Because target-capital movements are smaller, the gains from adjustment are smaller, and fewer plants switch adjustment status. Thus, the extensive margin no longer amplifies aggregate shocks.

6.4 Result 4: Nonconvex costs can eliminate spikes rather than create them

With volatile idiosyncratic productivity, plants may want to make large investment adjustments even without fixed costs. In this setting, fixed costs prevent some plants from adjusting. Therefore, nonconvex costs reduce the frequency of spikes relative to a frictionless model with idiosyncratic shocks.

6.5 Result 5: Partial-equilibrium estimation may overstate adjustment costs

The paper shows that plant-level investment dynamics differ substantially depending on whether general-equilibrium feedback is included. Since partial-equilibrium models attribute too much of the muted plant response to adjustment frictions, they can exaggerate the size of fixed costs.

7 How to Read the Tables and Figures

7.1 Table I: Baseline Parameter Values

Read:

- The table lists the baseline parameters used across the quantitative experiments: growth, discounting, depreciation, production elasticities, preference-for-leisure, aggregate productivity persistence and volatility, idiosyncratic productivity persistence and volatility, the small-adjustment band, and the upper support of the fixed cost distribution.
- Parameters not directly tied to adjustment costs are chosen to match long-run U.S. aggregate facts in the frictionless benchmark.

Economic message: The calibration first pins down standard aggregate features, then uses micro investment facts to discipline plant-specific shocks and adjustment frictions.

7.2 Table II: Distribution of Plant Investment Rates

Read:

- The table compares investment-rate bins in the data and the model, including inaction, small positive or negative investment, positive spikes, and negative spikes.
- The italicized empirical row reports the plant-level investment distribution that motivates the calibration.
- The lumpy model is designed to match the data closely, while a frictionless model with idiosyncratic shocks generates too many large adjustments.

Economic message: Micro-level realism requires both idiosyncratic productivity and a realistic treatment of small investment. Fixed costs are not simply a spike-generating device; they also suppress excessive spikes.

7.3 Table III: Role of Nonconvexities in Aggregate Investment Rate Dynamics

Read:

- Panel A reports partial-equilibrium moments. In partial equilibrium, the lumpy model generates much more skewness and excess kurtosis than the frictionless model.

- Panel B reports general-equilibrium moments. Once wages and interest rates clear markets, the lumpy and frictionless models look much more similar.

Economic message: Nonconvexities have large macro effects only when equilibrium prices are held fixed. General equilibrium absorbs the nonlinear response.

7.4 Figures 1 and 2: Partial-Equilibrium Aggregate Dynamics

Read:

- Figure 1 plots the distribution of aggregate investment rates in partial equilibrium. The lumpy model displays a more skewed and fat-tailed distribution than the frictionless model.
- Figure 2 traces aggregate capital responses after positive and negative productivity shocks. The asymmetric response helps explain the skewness in Figure 1.

Economic message: With fixed prices, lumpy adjustment creates aggregate nonlinearities because shocks change both the size of adjustments and the number of plants adjusting.

7.5 Figures 3 and 4: Adjustment Hazards and Extensive-Margin Responses

Read:

- Figure 3 shows the steady-state adjustment hazard in the common-productivity lumpy model. Plants far from target capital are more likely to adjust.
- Figure 4 shows how the adjustment hazard shifts after aggregate shocks. Positive shocks cause many plants to cross adjustment thresholds; negative shocks generate a different and weaker pattern.

Economic message: Aggregate nonlinearities arise from the interaction between the distribution of plants and the hazard of adjustment. The cross-sectional state distribution is therefore central to partial-equilibrium lumpy-investment dynamics.

7.6 Figure 5 and Table IV: General Equilibrium

Read:

- Figure 5 shows that the distribution of aggregate investment rates is much more compressed in general equilibrium than in partial equilibrium.
- Table IV compares aggregate business-cycle moments. The lumpy and frictionless models generate very similar cyclical behavior when solved in general equilibrium.

Economic message: Market-clearing movements in wages and interest rates dampen the target-capital response and remove most of the aggregate relevance of nonconvex plant adjustment costs.

7.7 Table V: Regression and Investment Moments

Read:

- The table reports moments from regressions and investment responses that help diagnose how aggregate shocks pass through the plant-level adjustment technology.
- The comparison reinforces that general-equilibrium dynamics differ sharply from partial-equilibrium dynamics even at the micro level.

Economic message: The difference between partial and general equilibrium is not just an aggregate accounting issue; it changes how individual plants respond to common shocks.

7.8 Table VI: Panel Regression Results

Read:

- The table regresses plant investment rates on aggregate and idiosyncratic productivity changes.
- The response to aggregate productivity shocks is much smaller in general equilibrium than in partial equilibrium.

Economic message: A researcher estimating adjustment frictions from panel data could misinterpret muted plant responses if equilibrium price feedback is ignored.

7.9 Table VII: Role of Fixed Costs in Plant-Level Investments

Read:

- The table compares models with different fixed-cost assumptions.
- When idiosyncratic productivity shocks are important, the model without fixed costs can generate too many spikes; fixed costs reduce them and help match the data.

Economic message: In a realistic heterogeneous economy, fixed costs are not necessarily the source of all lumpiness. They can also act as a filter that removes excessive idiosyncratic adjustments.

7.10 Tables A.I and A.II: Forecasting Rules

Read:

- These appendix tables report forecasting rules used in the numerical solution.
- They summarize how agents forecast aggregate state variables in the common-productivity and full lumpy models.

Economic message: The computation uses forecasting rules that allow the heterogeneous-agent economy to be solved recursively while tracking the aggregate distribution approximately.

8 Extracted Tables and Figures from the Paper

TABLE I
BASELINE PARAMETER VALUES

γ	β	δ	α	ν	φ	ρ_z	$\sigma_{\eta(z)}$	ρ_e	$\sigma_{\eta(e)}$	b	$\bar{\xi}$
1.016	0.077	0.060	0.756	0.640	2.400	0.950	0.014	0.950	0.077	0.011	0.0007

(a) Table I: Baseline parameter values

TABLE II
DISTRIBUTION OF PLANT INVESTMENT RATES

	Inaction	Positive Spike	Negative Spike	Positive Invest.	Negative Invest.
<i>Establishment data</i> ^a	0.081	0.186	0.018	0.815	0.104
Lumpy investment models ^b					
Traditional model	0.789	0.187	0.000	0.211	0.000
Extended model	0.073	0.185	0.010	0.752	0.175

^aData are from Cooper and Haltiwanger (2006). Inaction, $|i/k| < 0.01$; positive spike, $i/k > 0.20$; negative spike, $i/k < -0.20$; positive investment, $i/k \geq 0.01$; negative investment, $i/k \leq -0.01$.

^bThe traditional lumpy model has $\sigma_{e-} = b = 0$ and upper support = 0.014 to fit positive spike observations. The

(b) Table II: Distribution of plant investment rates

	Persistence	Standard Deviation	Skewness	Excess Kurtosis
<i>Postwar U.S. data</i> ^a	0.695	0.008	0.008	-0.715
A. Partial equilibrium models				
PE frictionless	-0.069	0.128	0.358	0.140
PE lumpy investment	0.210	0.085	1.121	2.313
B. General equilibrium models				
GE frictionless	0.659	0.010	0.048	0.048
GE lumpy investment	0.662	0.010	0.067	-0.074

^aData are annual private investment-to-capital ratio, 1954-2005, computed using Bureau of Economic Analysis tables.

the model and compare its results to those in an otherwise identical friction-

(a) Table III: Role of nonconvexities in aggregate dynamics

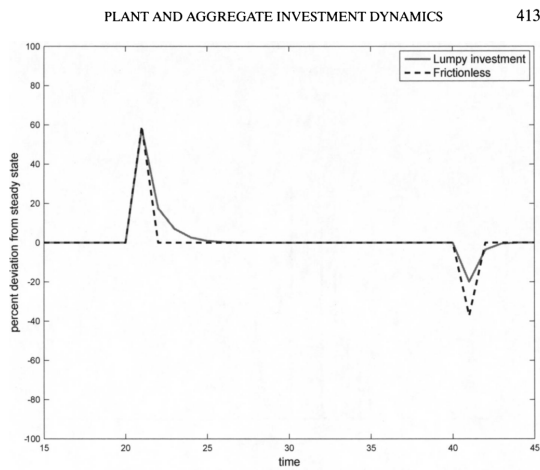


FIGURE 2.—Partial equilibrium aggregate capital responses.

(a) Figure 2: Partial-equilibrium capital responses

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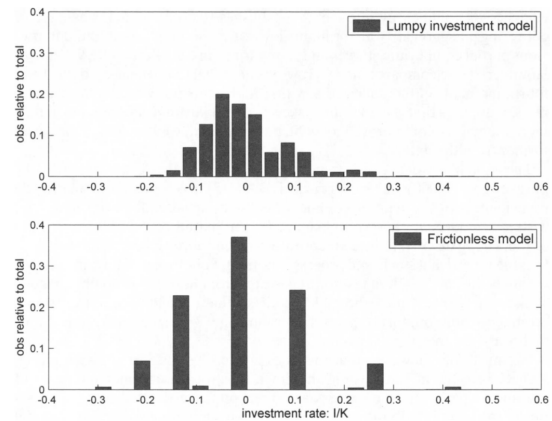


FIGURE 1.—Distribution of aggregate investment rates in partial equilibrium.

(b) Figure 1: Partial-equilibrium investment-rate distribution

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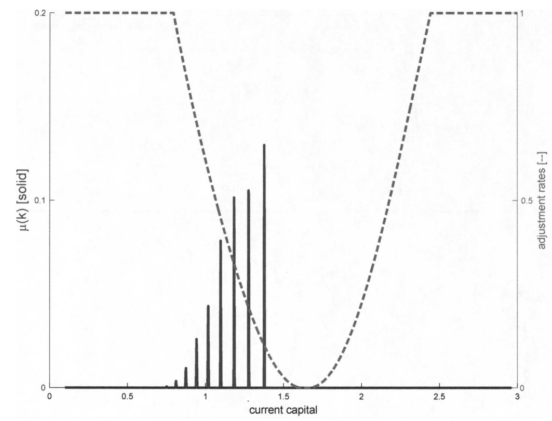


FIGURE 3.—Steady-state adjustment in the common productivity lumpy model.

(b) Figure 3: Steady-state adjustment

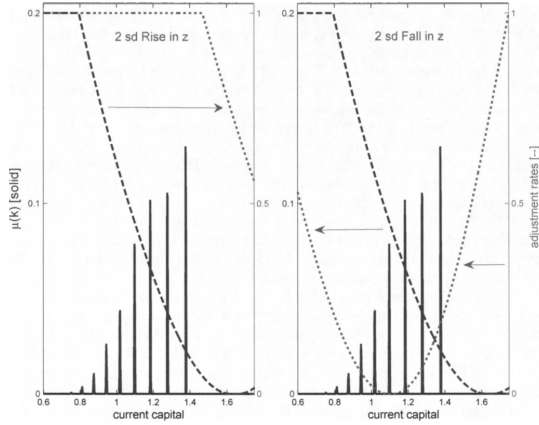


FIGURE 4.—Adjustment responses in the common productivity lumpy model.

(a) Figure 4: Adjustment responses

TABLE IV
AGGREGATE BUSINESS CYCLE MOMENTS

	Output	TFP ^a	Hours	Consump.	Invest.	Capital
A. Standard deviations relative to output ^b						
GE frictionless	(2.277)	0.602	0.645	0.429	3.562	0.494
GE lumpy	(2.264)	0.605	0.639	0.433	3.539	0.492
B. Contemporaneous correlations with output						
GE frictionless	1.000	0.955	0.895	0.976	0.034	
GE lumpy	1.000	0.956	0.900	0.976	0.034	

^aTotal factor productivity.^bThe logarithm of each series is Hodrick-Prescott-filtered using a weight of 100. The output column of panel A reports percent standard deviations of output in parentheses.

(a) Table IV: Aggregate business-cycle moments

TABLE VI
EQUILIBRIUM AND THE DYNAMICS OF PLANT INVESTMENT RATES

Panel Regression Results Using Data From Simulated Lumpy Investment Model	Constant	Δz	$\Delta \epsilon$	Adjusted R-Squared	S.E. of Regression
General equilibrium	0.107	0.423 (0.047)	5.126 (0.032)	0.227	0.215
Partial equilibrium	0.112	5.464 (0.051)	5.414 (0.034)	0.289	0.233

The sharply differing microlevel elasticity to changes in aggregate total fac-

(a) Table VI: Panel regression results

These observations are confirmed by the second and higher moments presented in Table III. There we see an unambiguous improvement in model fit as we move from the partial equilibrium (PE) lumpy investment row of panel A to its general equilibrium (GE) counterpart in panel B. Persistence increases sharply, nearly reaching the empirical autocorrelation, while the excessive volatility in the adjacent column is virtually eliminated. Moreover, comparing the GE lumpy model to its PE counterpart, we see that equilibrium dramatically reduces the skewness and excess kurtosis in the distribution of ag-

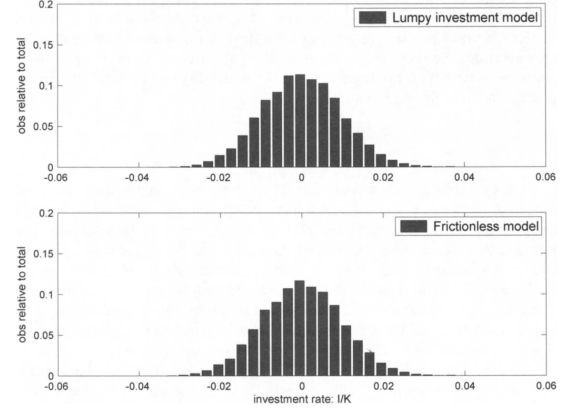


FIGURE 5.—Distribution of aggregate investment rates in general equilibrium.

(b) Figure 5: General-equilibrium investment-rate distribution

TABLE V
ROBUSTNESS

	Persistence	Standard Deviation	Skewness	Excess Kurtosis
Postwar U.S. data	0.706	0.008	0.008	-0.715
Baseline lumpy investment model	0.662	0.010	0.067	-0.074
Modified versions of baseline model ^a				
1. Big fixed costs	0.677	0.009	0.071	-0.066

(b) Table V: Regression and investment moments

TABLE VII
ROLE OF FIXED COSTS IN PLANT-LEVEL INVESTMENTS

	Inaction	Positive Spike	Negative Spike	Positive Invest.	Negative Invest.
Establishment data ^a	0.081	0.186	0.018	0.815	0.104
Models with plant-specific TFP shocks					
1. Frictionless model	0.032	0.204	0.028	0.611	0.356
2. Lumpy investment model	0.073	0.185	0.010	0.752	0.175

^aData are from Cooper and Haltiwanger (2006). Inaction, $|i/k| < 0.01$; positive spike, $i/k > 0.20$; negative spike, $i/k < -0.20$; positive investment, $i/k \geq 0.01$; negative investment, $i/k \leq -0.01$.

(b) Table VII: Role of fixed costs in plant-level investments

TABLE A.I
FORECASTING RULES IN COMMON PRODUCTIVITY LUMPY MODEL

Productivity ^a	β_0	β_1	S.E.	Adj. R^2
A. Forecasting m'_1				
z_1 (119 obs)	0.006	0.802	0.20e-3	0.9999
z_2 (298 obs)	0.014	0.800	0.25e-3	0.9999
z_3 (734 obs)	0.021	0.798	0.25e-3	0.9999
z_4 (1,208 obs)	0.028	0.797	0.28e-3	0.9999
z_5 (1,682 obs)	0.035	0.796	0.31e-3	0.9999
z_6 (1,871 obs)	0.041	0.796	0.31e-3	0.9999
z_7 (1,706 obs)	0.048	0.795	0.29e-3	0.9999
z_8 (1,237 obs)	0.055	0.794	0.26e-3	0.9999
z_9 (751 obs)	0.062	0.793	0.24e-3	0.9999
z_{10} (295 obs)	0.070	0.791	0.26e-3	0.9999
z_{11} (99 obs)	0.078	0.789	0.21e-3	0.9999
B. Forecasting p				
z_1 (119 obs)	0.998	-0.395	0.14e-3	0.9999
z_2 (298 obs)	0.990	-0.393	0.15e-3	0.9999
z_3 (734 obs)	0.981	-0.392	0.14e-3	0.9999
z_4 (1,208 obs)	0.972	-0.392	0.15e-3	0.9999
z_5 (1,682 obs)	0.963	-0.391	0.15e-3	0.9999
z_6 (1,871 obs)	0.953	-0.389	0.15e-3	0.9999
z_7 (1,706 obs)	0.944	-0.388	0.14e-3	0.9999
z_8 (1,237 obs)	0.935	-0.386	0.13e-3	0.9999
z_9 (751 obs)	0.926	-0.386	0.11e-3	0.9999
z_{10} (295 obs)	0.917	-0.385	0.11e-3	0.9999
z_{11} (99 obs)	0.908	-0.384	0.11e-3	0.9999

^aForecasting rules are conditional on current aggregate total factor productivity z_t . Each regression takes the form $\log(y) = \beta_0 + \beta_1 \log(m_1)$, where $y = m'_1$ or p .

(a) Table A.I: Forecasting rules in common-productivity lumpy model

TABLE A.II
FORECASTING RULES IN FULL LUMPY MODEL

Productivity ^a	β_0	β_1	S.E.	Adj. R^2
A. Forecasting m'_1				
z_1 (119 obs)	0.009	0.800	0.15e-3	1.0000
z_2 (298 obs)	0.016	0.798	0.22e-3	0.9999
z_3 (734 obs)	0.023	0.796	0.23e-3	0.9999
z_4 (1,208 obs)	0.030	0.795	0.26e-3	0.9999
z_5 (1,682 obs)	0.037	0.794	0.27e-3	0.9999
z_6 (1,871 obs)	0.044	0.793	0.28e-3	0.9999
z_7 (1,706 obs)	0.051	0.793	0.26e-3	0.9999
z_8 (1,237 obs)	0.058	0.792	0.24e-3	0.9999
z_9 (751 obs)	0.065	0.792	0.23e-3	0.9999
z_{10} (295 obs)	0.072	0.791	0.25e-3	0.9999
z_{11} (99 obs)	0.079	0.791	0.19e-3	0.9999
B. Forecasting p				
z_1 (119 obs)	0.994	-0.397	0.03e-3	1.0000
z_2 (298 obs)	0.986	-0.395	0.04e-3	1.0000
z_3 (734 obs)	0.977	-0.394	0.04e-3	1.0000
z_4 (1,208 obs)	0.968	-0.393	0.05e-3	1.0000
z_5 (1,682 obs)	0.958	-0.392	0.05e-3	1.0000
z_6 (1,871 obs)	0.949	-0.391	0.05e-3	1.0000
z_7 (1,706 obs)	0.940	-0.389	0.05e-3	1.0000
z_8 (1,237 obs)	0.931	-0.388	0.05e-3	1.0000
z_9 (751 obs)	0.921	-0.386	0.04e-3	1.0000
z_{10} (295 obs)	0.912	-0.384	0.05e-3	1.0000
z_{11} (99 obs)	0.903	-0.382	0.04e-3	1.0000

^aForecasting rules are conditional on current aggregate total factor productivity z_t . Each regression takes the form $\log(y) = \beta_0 + \beta_1 \log(m_1)$, where $y = m'_1$ or p .

(b) Table A.II: Forecasting rules in full lumpy model

9 Research Design and Identification Logic

This paper is primarily a quantitative structural exercise rather than a reduced-form empirical identification paper. The identification logic is therefore based on disciplined calibration, model comparison, and counterfactual experiments.

What is disciplined by data?

- Aggregate parameters are chosen to match long-run U.S. macro moments.
- The distribution of plant investment rates disciplines the idiosyncratic productivity process and adjustment-cost parameters.
- Simulated aggregate moments are compared with postwar U.S. business-cycle moments.

What is identified through model comparison?

- Comparing lumpy versus frictionless models isolates the role of nonconvex adjustment costs.
- Comparing partial versus general equilibrium isolates the role of market-clearing price feedback.
- Comparing common-productivity versus full heterogeneous models isolates the role of idiosyncratic productivity risk.

10 Key Mechanism in One Chain

The core mechanism can be summarized as follows:

Aggregate productivity shock $\rightarrow$ target capital changes $\rightarrow$ adjustment value changes $\rightarrow$ adjustment hazard change

In partial equilibrium, the first arrow is strong because wages and interest rates do not move. In general equilibrium, the first arrow is weak because prices move to clear markets. Once target-capital movements are small, the extensive-margin response also becomes small.

11 Why the Result Matters

- **For macroeconomics:** Standard smooth representative-agent models may approximate aggregate investment dynamics better than plant-level lumpiness alone would suggest.
- **For investment theory:** The effect of nonconvex adjustment costs depends on the size and persistence of idiosyncratic shocks.
- **For empirical work:** Estimating adjustment costs without equilibrium feedback may lead to biased conclusions.
- **For heterogeneous-agent modeling:** Matching micro distributions is necessary but not sufficient; one must also ask whether micro frictions survive aggregation and equilibrium feedback.

12 Limitations and Discussion

- The model is highly structured and relies on calibration rather than direct estimation of all primitives.
- The annual frequency is useful for matching plant-level investment evidence but may miss higher-frequency adjustment behavior.
- The representative-household equilibrium structure generates strong price feedback; alternative financial or labor-market frictions could weaken this smoothing mechanism.
- The model focuses on capital adjustment and does not directly model financing constraints, credit frictions, or firm entry and exit in the way later heterogeneous-firm macro models often do.

13 Takeaway

The paper's main lesson is that plant-level nonconvexities do not automatically translate into aggregate nonlinearities. In partial equilibrium, lumpy investment creates large changes in aggregate investment through the extensive margin of plant adjustment. In general equilibrium, however, wage and interest-rate movements dampen target-capital changes, reduce adjustment incentives, and make aggregate investment dynamics close to those of a frictionless model. The paper therefore shifts the question from whether plant investment is lumpy to whether that lumpiness survives equilibrium feedback.